

Conditioning on the Largest Color Classes: Certified Bounds for Minimum-Sum Coloring of Very Dense Benchmark Graphs

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Abstract

In the minimum-sum coloring problem the cost of a coloring after canonical relabeling depends only on the sizes of its color classes, so lower bounds can be obtained by optimizing over class-size profiles. Such bounds do not see whether stable sets of the required sizes can coexist. We strengthen them by conditioning on which maximum stable sets are color classes and then bounding, layer by layer, the stable sets that can still be packed into the remaining vertices. Each bound is certified by exact rational dual inequalities, combined where necessary by exhaustive branching, over stable-set families enumerated completely from the graph.

For the DIMACS graph DSJC250.9 this proves that the chromatic sum is 8,277, the value of colorings reported in the literature; the lower bound previously reported was 7,882. We also prove $29,791 \leq \Sigma(\text{DSJC500.9}) \leq 29,848$ and $102,344 \leq \Sigma(\text{DSJC1000.9}) \leq 103,256$, raising the previously reported lower bounds of 29,768 and 100,078, and $381,823 \leq \Sigma(\text{C2000.9}) \leq 382,379$. For C2000.9 the lower bound coincides with the optimum of the stable-set linear relaxation, and for DSJC500.9 the three lowest remaining cases admit fractional solutions of the conditioned relaxation, so raising either lower bound requires arguments beyond that relaxation. All four graphs have independence number 5 or 6, which is what makes the complete enumerations possible.

1 Introduction

In the minimum-sum coloring problem (MSCP), introduced by Kubicka and Schwenk [11], the vertices of a graph receive positive integer colors, adjacent vertices receive different colors, and the sum of the colors is to be minimized. The minimum is the *chromatic sum* $\Sigma(G)$. Every vertex pays the label of its class, so in an optimal coloring larger classes receive smaller labels, and the cost is determined by the sorted sizes of the classes. The number of colors enters only through those sizes: a coloring with more than $\chi(G)$ colors can have a smaller sum, because the gain from a large class at a small label can outweigh the cost of an extra small class at the end.

Lower bounds for the problem have developed along two lines. The first discards part of the graph's structure. A partition of the vertices into cliques gives a bound, since a clique of size k needs k distinct labels [16, 22]; it keeps the incompatibilities inside each clique and discards the edges between cliques. Profile relaxations go further and keep only the sizes of the classes, constrained by a few graph invariants. Lecat, Lucet and Li [13] minimize the sorted-profile cost over integer partitions of n whose parts are at most $\alpha(G)$ and whose number of parts is at least a lower bound on $\chi(G)$. Gondran, Duchamp and Moalic [7] add a cap on the number of parts equal to $\alpha(G)$, computed from the intersection graph of the maximum stable sets; without the cap, their relaxation reduces to that of Lecat et al. The second line keeps the graph intact through stable-set formulations, in which a variable selects each possible color class. Mehrotra and Trick [15] showed how to solve this formulation of ordinary coloring by column generation. It has been adapted to MSCP by Furini et al. [5] and Mrad et al. [18], and used in a branch-and-price

Table 1: Bounds for the four graphs. The reported values are taken from the tables cited in the notes; their proofs and colorings were not replayed here. The bounds of this paper are proved in the sections indicated, and the gap is the difference between them.

graph	n	α	reported		this paper			section
			lower	upper	lower	upper	gap	
DSJC250.9	250	5	7,882 ^a	8,277 ^b	8,277	8,277	0	4
DSJC500.9	500	5	29,768 ^a	29,856 ^c	29,791	29,848	57	5
DSJC1000.9	1,000	6	100,078 ^a	103,429 ^c	102,344	103,256	912	6.1
C2000.9	2,000	6	— ^d	— ^d	381,823	382,379	556	6.2

^a Gondran et al. [7, Table 1] (preprint). ^b Lecat et al. [13, Table 2]; also reported in [9, 17]. ^c EBES [23, Table 1], five-hour runs. ^d No MSCP values were located in the principal heuristic tables of [9, 17, 21, 23].

algorithm by Delle Donne et al. [3]. Tammal and Aïder [20] give an exact method based instead on modular decomposition.

What profile relaxations cannot see is simultaneous compatibility. A profile may ask for many large classes even though every choice of those classes leaves too little room for the smaller ones. The cap of Gondran et al. records how many maximum classes can coexist, but not what a particular choice of them leaves behind, and two maximum stable sets of the same size can leave very different room below them; Figure 1 shows the smallest kind of example. This paper conditions on the identities of the largest color classes and then bounds, one layer at a time, how many stable sets of each smaller size can still be packed. The bounds are certified by exact rational dual inequalities, combined, when one inequality is not enough, into trees of inequalities obtained by exhaustive branching. They are checked against families of stable sets enumerated completely from the graph.

Complete enumeration is possible because the graphs we study are very dense. Held, Cook and Sewell [8] note that in graphs of edge density about 0.9 the maximal stable sets are few and small enough to enumerate; their safe branch-and-price settled the chromatic number of DSJC250.9, which is 72 [8, Table 3]. We work with four DIMACS graphs of this density, DSJC250.9, DSJC500.9, DSJC1000.9 and C2000.9, whose independence numbers are 5 or 6.

1.1 Results

Theorem 1.1. *The chromatic sums of the four benchmark graphs¹ satisfy*

- (i) $\Sigma(\text{DSJC250.9}) = 8,277$;
- (ii) $29,791 \leq \Sigma(\text{DSJC500.9}) \leq 29,848$;
- (iii) $102,344 \leq \Sigma(\text{DSJC1000.9}) \leq 103,256$;
- (iv) $381,823 \leq \Sigma(\text{C2000.9}) \leq 382,379$.

Every upper bound comes from an explicit coloring checked against the graph. The colorings were found by the author’s coloring searches, followed for C2000.9 by the reoptimization described in Section 6.2. Since each is checked directly, the bounds do not depend on how the colorings were found. Table 1 compares the bounds with those reported in the literature.

For DSJC250.9 the reported lower bounds rose from 4,179 by clique decomposition [22, Table 1] to 6,651 with motifs [13, Table 2] and 7,882 with the cap on maximum classes [7, Table 1], while colorings of sum 8,277 were reported in [9, 13, 17]; the EXSCOL heuristic reported 8,286 [21, Table 1]. Theorem 1.1(i) closes this gap. For DSJC500.9 and DSJC1000.9 our lower bounds exceed those reported by Gondran et al. by 23 and 2,266, and our colorings improve the EBES values by 8 and 173. Of the 2,266 gained on DSJC1000.9, a feasible dual of the

¹The graph files, and the identities that bind the theorem to them, are specified in Appendix E.

unconditioned stable-set relaxation already accounts for 2,231, since its value rounds up to 102,309; conditioning on the three stable 6-sets adds the remaining 35 (Section 6.1). For C2000.9 the lower bound is the profile bound of Gondran et al. evaluated with a certified cap, and the stable-set relaxation attains it exactly (Section 6.2).

1.2 What is inherited and what is new

Several ingredients come from earlier work. The partition relaxations that work with sorted class sizes are those of Lecat et al. [13] and Gondran et al. [7]. We write profiles in Ferrers (conjugate-partition) coordinates [1], which also appear in integer-programming formulations [6], because in these coordinates the objective becomes a sum of triangular numbers. The cap on the number of maximum classes is that of Gondran et al.; we compute its value exactly and certify it. The stable-set master of Section 2.3 is a stable-set formulation in the tradition of Mehrotra and Trick, with the sum objective written in Ferrers coordinates. Our lower bounds rest, like the safe bounds of Held, Cook and Sewell [8], on dual solutions with integer data whose feasibility is verified exactly, and our branch trees are checked in the spirit of certified branch-and-bound [2, 12].

The contribution of this paper is the conditioning and its certification: fixing the identities of the top classes, bounding the layers below them by lexicographic scores, and certifying each bound by exact dual inequalities and exhaustively branched trees over complete residual stable-set families (Section 3). With these tools we prove the bounds of Theorem 1.1. The same analysis locates where the approach stops. The three lowest open cases for DSJC500.9 admit fractional solutions of the conditioned relaxation (Section 5.4), and for C2000.9 the profile bound already equals the optimum of the stable-set relaxation (Section 6.2).

1.3 Organization

Section 2 develops profiles, Ferrers coordinates and the stable-set master. Section 3 introduces top packings and envelopes and the certificates that bound them. Section 4 proves $\Sigma(\text{DSJC250.9}) = 8,277$, Section 5 proves the bound for DSJC500.9 and shows why its three lowest cases resist, and Section 6 conditions the stable-set relaxation for DSJC1000.9 and evaluates it for C2000.9. Section 7 explains how the results are checked, and Section 8 discusses what the method needs and what remains open. The appendices hold the proof data for DSJC250.9 (Appendices A and B), two supplementary analyses that no proof depends on (Appendices C and D), the file identities and replay commands (Appendix E), and a table of notation (Appendix F).

2 Profiles and the stable-set master

2.1 Class-size profiles and Ferrers coordinates

Let $G = (V, E)$ be a finite simple graph with $n = |V|$. A proper sum coloring is a map $c : V \rightarrow \mathbb{Z}_{>0}$ with $c(u) \neq c(v)$ for every $uv \in E$; its value is $\sum_{v \in V} c(v)$, and $\Sigma(G)$ is the least value of a proper sum coloring. The nonempty color classes are stable sets. Deleting unused labels and compressing the others to $1, \dots, \ell$ cannot increase the sum, so we write the classes as $C_1, \dots, C_\ell$, with the subscript as color.

For a fixed partition into stable classes, consecutive labels should be assigned in nonincreasing order of class size: if $i < j$ but $|C_i| < |C_j|$, exchanging the two labels changes the objective by

$$i|C_j| + j|C_i| - i|C_i| - j|C_j| = (j - i)(|C_i| - |C_j|) < 0.$$

Every coloring therefore has a canonical relabeling whose class sizes $\lambda = (\lambda_1, \dots, \lambda_\ell)$ satisfy $\lambda_1 \geq \dots \geq \lambda_\ell \geq 1$. We call λ the sorted class-size profile, and call a profile *feasible* when V can be partitioned into stable sets of exactly those sizes.

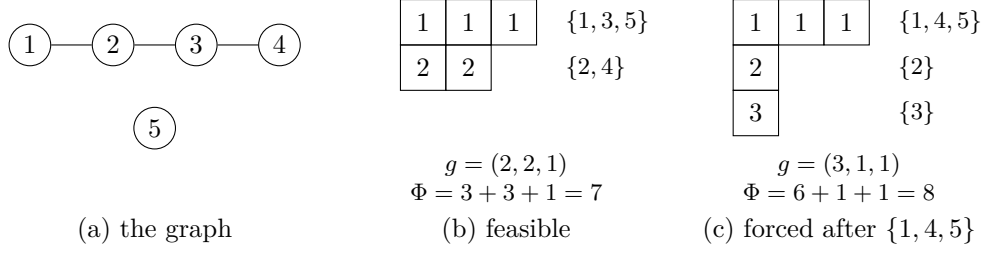


Figure 1: Profiles forget which vertices form the classes. For the graph (a), each Ferrers diagram has one row per class, labelled by its color, so the row sums give the objective class by class and the column sums are the triangular numbers $T(g_t)$. The profile $(3, 2)$ is realized with the size-3 class $\{1, 3, 5\}$ (b), but not with the size-3 class $\{1, 4, 5\}$, which leaves the adjacent vertices 2 and 3 as singletons (c).

Let $\alpha = \alpha(G)$ be the independence number. For $1 \leq s \leq \alpha$ let $h_s = |\{i : \lambda_i = s\}|$ be the number of classes of size exactly s , and for $1 \leq t \leq \alpha$ let

$$g_t = \sum_{s=t}^{\alpha} h_s = |\{i : \lambda_i \geq t\}|$$

be the number of classes of size at least t . The vector g is the conjugate of the partition λ [1]; we call its entries the *Ferrers coordinates*. They satisfy $g_1 = \ell$ and $g_1 \geq g_2 \geq \dots \geq g_{\alpha} \geq 0$, and $h_s = g_s - g_{s+1}$ with $g_{\alpha+1} = 0$. Throughout, $T(q) = q(q+1)/2$ denotes the q th triangular number.

Proposition 2.1 (Ferrers identities). *For every sorted class-size profile,*

$$n = \sum_{s=1}^{\alpha} s h_s = \sum_{t=1}^{\alpha} g_t, \tag{1}$$

$$\sum_{i=1}^{\ell} i \lambda_i = \sum_{t=1}^{\alpha} T(g_t). \tag{2}$$

Proof. A class of size s contributes one cell to each of the first s columns of the Ferrers diagram, which proves (1). For the objective,

$$\sum_{i=1}^{\ell} i \lambda_i = \sum_{i=1}^{\ell} i \sum_{t=1}^{\lambda_i} 1 = \sum_{t=1}^{\alpha} \sum_{i=1}^{g_t} i = \sum_{t=1}^{\alpha} T(g_t). \quad \square$$

We write $\Phi(h) = \sum_{t=1}^{\alpha} T(g_t)$ for the cost determined by the exact-size profile h .

Corollary 2.2 (Profile lower bound). *Every proper coloring whose sorted exact-size profile is h has value at least $\Phi(h)$, with equality when consecutive labels are assigned in nonincreasing order of class size.*

Proof. Compress unused labels, apply the exchange argument, and use Proposition 2.1. $\square$

Figure 1 shows what a profile forgets. The graph has vertices $1, \dots, 5$ and edges $12, 23, 34$. Its maximum stable sets are $\{1, 3, 5\}$, $\{1, 4, 5\}$ and $\{2, 4, 5\}$. With the classes $\{1, 3, 5\}$ and $\{2, 4\}$ the profile is $\lambda = (3, 2)$, with $h = (0, 1, 1)$, $g = (2, 2, 1)$ and $\Phi(h) = T(2) + T(2) + T(1) = 7$; the maximum class $\{2, 4, 5\}$ also admits it, with the pair $\{1, 3\}$. If the maximum class is $\{1, 4, 5\}$, however, the remaining vertices 2 and 3 are adjacent, the profile becomes $(3, 1, 1)$ and the sum is 8. A bound that records only how many classes of size three can occur cannot tell these cases apart. A bound that conditions on the identity of the size-three class can, and that is the device developed in Section 3.

2.2 Feasibility as a packing problem

Let $\mathcal{S}_s$ be the family of stable sets of size s in G . Singletons impose no compatibility condition, so they can be removed from the combinatorial core of the problem.

Proposition 2.3 (Singleton completion). *A nonnegative integer profile $h = (h_1, \dots, h_\alpha)$ with $\sum_s s h_s = n$ is feasible if and only if there are pairwise-disjoint stable sets consisting of exactly h_s members of $\mathcal{S}_s$ for each $s \geq 2$. The uncovered vertices then form the singleton classes, and there are exactly h_1 of them.*

Proof. The classes of size at least two in a coloring with profile h form such a packing. Conversely, a packing with these counts covers $\sum_{s \geq 2} s h_s$ vertices, so by (1) it leaves h_1 uncovered vertices, each a stable singleton. $\square$

Feasibility of h is thus the typed set-packing system

$$\sum_{Q \in \mathcal{S}_s} x_Q = h_s \quad (s = 2, \dots, \alpha), \quad \sum_{Q \ni v} x_Q \leq 1 \quad (v \in V), \quad x_Q \in \{0, 1\}.$$

Relaxing an equality to an upper bound enlarges the set of packings, so an upper bound on packings proved under the relaxed counts remains valid; we use this repeatedly.

2.3 The stable-set master

Let $\mathcal{S}$ be the family of all nonempty stable sets of G . The integer master has a binary variable x_S for each $S \in \mathcal{S}$ and the exact-cover constraints

$$\sum_{S \ni v} x_S = 1 \quad (v \in V). \quad (3)$$

For each threshold t put $q_t = \sum_{S: |S| \geq t} x_S$. An integral solution is a partition of V into stable classes, $q_t = g_t$ counts its classes of size at least t , and by Proposition 2.1 its cost after canonical relabeling is $\sum_t T(q_t)$. Conversely, every proper coloring gives an integral solution. The integer master is therefore equivalent to MSCP.

To obtain a linear model, put $N_t = \lfloor n/t \rfloor$ and introduce unit cells $0 \leq u_{t,j} \leq 1$ for $1 \leq j \leq N_t$:

$$\sum_{j=1}^{N_t} u_{t,j} = q_t \quad (1 \leq t \leq \alpha), \quad \text{minimize} \quad \sum_{t=1}^{\alpha} \sum_{j=1}^{N_t} j u_{t,j}. \quad (4)$$

For integral q_t the cheapest q_t cells are filled and the row costs exactly $T(q_t)$. For fractional $q_t \in [0, N_t]$ it costs $T(\lfloor q_t \rfloor) + (\lfloor q_t \rfloor + 1)(q_t - \lfloor q_t \rfloor)$, the piecewise-linear interpolation of T rather than $q_t(q_t + 1)/2$. Relaxing x_S to $0 \leq x_S \leq 1$ gives the *master relaxation*, in which the coordinates q_t may be fractional; its value, rounded up, is a lower bound on $\Sigma(G)$.

This is a stable-set formulation in the tradition of Mehrotra and Trick [15]. Delle Donne et al. [3] give each pair of a stable set and a color its own variable; here the colors are aggregated into Ferrers thresholds and the triangular cost is linearized by unit cells. Two facts are used throughout. A rational dual solution that is feasible for every column certifies a lower bound on the relaxation by weak duality, and a rational primal solution of the same value certifies the optimum. A floating-point objective or a solver's status carries neither implication.

For DSJC1000.9 and C2000.9 we bound or compute the optimum of this relaxation directly (Section 6); for DSJC500.9 we show that its conditioned form cannot exclude the lowest remaining cases (Section 5.4). The conditional arguments of Sections 3 to 5 bound integer packings directly.

3 Conditioning on the largest classes

Throughout, B is one less than the value of a known coloring. To prove that the coloring is optimal, or to bound how far it is from optimal, we exclude colorings of value at most B . Profiles of value exactly B must still be excluded, so a bound removes a profile only when it is strictly greater than B . Choosing B from a known coloring assumes nothing about optimality.

3.1 Top packings, cells and envelopes

Definition 3.1. A *top packing* is a family F of pairwise-disjoint stable sets of size α . A coloring *has top packing* F if its classes of size α are exactly the members of F . Every coloring has exactly one top packing, which may be empty; other stable α -sets may exist in $G - \bigcup F$, but in a coloring with top packing F none of them is a color class. A *cell* is a pair (F, h) of a top packing and a profile with $h_\alpha = |F|$.

Fix a top packing F and a set $E \subseteq \{2, \dots, \alpha - 1\}$ of *eligible sizes*. The *complete residual universe* is

$$\mathcal{Q}_F(E) = \left\{ Q \subseteq V \setminus \bigcup F : Q \text{ is stable and } |Q| \in E \right\},$$

and a *residual packing* is a pairwise-disjoint subfamily $\mathcal{A} \subseteq \mathcal{Q}_F(E)$, with $n_s(\mathcal{A})$ members of size s . Given prescribed numbers of residual classes of each size above s , the *conditional envelope* at size s is the largest number of s -sets in a residual packing with those numbers.

A cell whose profile asks for more s -classes than the envelope allows is infeasible. The next proposition states the form in which we use this. It allows several sizes to be scored at once, which is how envelopes given counts at larger sizes are certified (Section 3.3).

Proposition 3.2 (Conditional exclusion). *Let F be a top packing, E a set of eligible sizes with scores $a_s > 0$, c_s caps on some of the sizes in E , and τ an integer. Suppose every residual packing $\mathcal{A} \subseteq \mathcal{Q}_F(E)$ with $n_s(\mathcal{A}) \leq c_s$ for each capped size satisfies $\sum_{Q \in \mathcal{A}} a_{|Q|} < \tau$. Then no coloring with top packing F has residual class counts n_s ($s \in E$) with $n_s \leq c_s$ for each capped size and $\sum_{s \in E} a_s n_s \geq \tau$.*

Proof. The classes of such a coloring whose sizes lie in E are stable, pairwise disjoint and disjoint from $\bigcup F$, so they lie in $\mathcal{Q}_F(E)$ and form a residual packing that respects the caps and scores at least τ , a contradiction. Classes of other sizes are simply ignored, which can only enlarge the family of packings considered. $\square$

The hypothesis that $\mathcal{Q}_F(E)$ contains *every* eligible stable set is essential: a coloring could otherwise use a class that the model omits. This is why every residual universe in this paper is regenerated from the graph rather than supplied as a list.

3.2 Dual certificates

Packing bounds are certified by the following form of weak duality, written with integer data so that it can be checked without rounding.

Lemma 3.3 (Scaled dual bound). *Let $\mathcal{Q}$ be a family of stable sets with scores $a_s > 0$ for the sizes that occur in it. For each size s let the number $n_s(\mathcal{A})$ of members of size s in a subfamily $\mathcal{A}$ be either unrestricted, bounded ($n_s \leq c_s$) or fixed ($n_s = c_s$). Let D be a positive integer, $y_v \geq 0$ integer vertex weights, and μ_s integers with $\mu_s = 0$ for unrestricted sizes, $\mu_s \geq 0$ for bounded sizes and μ_s of either sign for fixed sizes. If*

$$\sum_{v \in Q} y_v + \mu_{|Q|} \geq D a_{|Q|} \quad (Q \in \mathcal{Q}), \tag{5}$$

then every pairwise-disjoint subfamily $\mathcal{A} \subseteq \mathcal{Q}$ that meets the count conditions satisfies

$$D \sum_{Q \in \mathcal{A}} a_{|Q|} \leq \sum_v y_v + \sum_s \mu_s c_s. \quad (6)$$

Proof. Sum (5) over $\mathcal{A}$. By disjointness each nonnegative weight y_v is counted at most once. The multiplier terms sum to $\sum_s \mu_s n_s(\mathcal{A})$; on fixed sizes this equals $\mu_s c_s$, on bounded sizes it is at most $\mu_s c_s$ because $\mu_s \geq 0$, and on unrestricted sizes it vanishes. $\square$

The scores and count bounds used here are integers. Thus (5) is checked set by set and (6) is evaluated exactly; any floor is taken only after the scaled inequality has been established. With unit scores on the sizes at least t , the left side of (6) is D times the number of classes of size at least t . For example, the global 4-set certificate for DSJC250.9 (S1 in [Appendix A](#)) has $D = 26$ and $\sum_v y_v = 1039$ and no count multipliers. It is checked against all 205 stable 4-sets and all three stable 5-sets of the graph, and it gives $26g_4 \leq 1039$, hence $g_4 \leq \lfloor 1039/26 \rfloor = 39$.

This is the same principle as the safe column generation of Held, Cook and Sewell [8], who scale a floating-point dual solution to integers and certify the scaled solution by an exact computation. In our setting the certification is a direct check against every column of an enumerated family.

When some columns have already been selected, the lemma gives a test for closing a case.

Corollary 3.4 (Leaf test). *Suppose columns of total score p have been selected, let $\mathcal{Q}'$ be the columns disjoint from them that remain available, and let c'_s be the remaining caps. If a certificate for $\mathcal{Q}'$ and c' as in [Lemma 3.3](#), with $\mu_s \geq 0$ for capped and $\mu_s = 0$ for uncapped sizes, satisfies*

$$Dp + \sum_v y_v + \sum_s c'_s \mu_s < D\tau, \quad (7)$$

then no packing that extends the selection reaches score τ .

The strict inequality (7) is checked as written, without division or rounding. A single certificate is the special case with nothing selected and $p = 0$.

3.3 Lexicographic scores

An envelope asks for the largest number of s -sets given the counts at larger sizes. We certify it with one score that ranks larger sizes first. Let $\rho = |V| + 1$. Every residual count is at most $|V| < \rho$, and $\rho^2 > |V|\rho + |V|$, so one unit in a higher coordinate outweighs any variation in the lower ones. Hence

$$Z_3 = \rho n_4 + n_3 \quad \text{and} \quad Z_2 = \rho^2 n_4 + \rho n_3 + n_2$$

order packings lexicographically by (n_4, n_3) and by (n_4, n_3, n_2) , exactly and without tolerances. For example, if every packing with $n_4 \leq 37$ has $Z_3 < 37\rho + 30$, then a packing with 37 residual 4-sets has at most 29 residual 3-sets. We use $\rho = 251$ for DSJC250.9 and $\rho = 501$ for DSJC500.9.

3.4 Branch trees

When one certificate is not strong enough, we branch. A node records the columns selected so far, their score p , the active columns (those disjoint from the selection and not excluded) and the remaining caps. An internal node branches on an active column Q . Its zero child excludes Q ; its one child selects Q , adds its score, decrements the remaining quota of $|Q|$ when that size is capped, and removes every active column that meets Q , which is valid because the other members of a packing containing Q are disjoint from it. A tree is *admissible* for a target τ if its root holds the complete residual universe with the model's caps and $p = 0$, every internal node branches on an active column and has both children, every node has valid nonnegative remaining quotas, and every leaf passes the test of [Corollary 3.4](#).

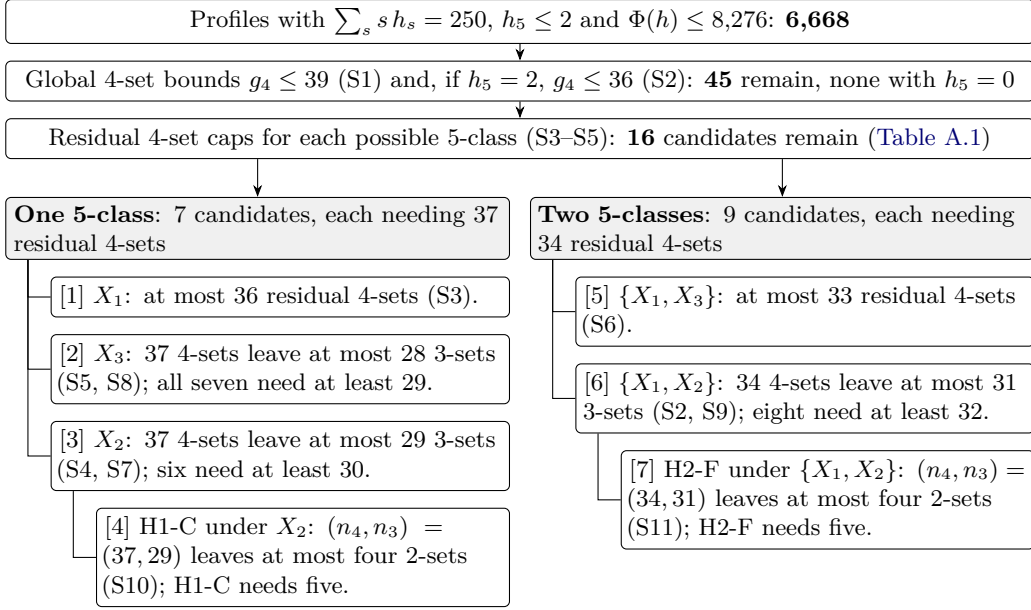


Figure 2: The proof that no coloring of DSJC250.9 has sum at most 8,276. Bracketed numbers are the branches of Table 4; S1–S11 are the certificates of Table A.2. X_1 , X_2 , X_3 are the three maximum stable sets.

Proposition 3.5 (Soundness of trees). *If a tree is admissible for the target τ , then no residual packing of its model reaches score τ .*

Proof. Every packing represented at an internal node either omits the branching column, and is represented at the zero child, or contains it, and is represented at the one child after the conflicting columns are removed. Each leaf excludes all packings it represents, so by induction from the leaves the root excludes every packing of the model. $\square$

The six DSJC250.9 trees have at most 1,661 nodes (Table A.3). General certificate formats for integer programming represent the same kind of case analysis differently. VIPR [2] flattens a branch-and-bound tree into a sequence of derived inequalities, introducing each branching disjunction as a pair of assumptions and discharging them by ‘unsplitting’; Larrosa et al. [12] separate branch-and-bound search from its verification through checkable reasons for each bound closure. Our trees need a single inference at the leaves, the bound of Lemma 3.3, because each leaf is closed by a packing dual checked against an explicit column family.

4 The chromatic sum of DSJC250.9

This section proves $\Sigma(\text{DSJC250.9}) = 8,277$. The argument has the shape shown in Figure 2. An improving coloring has at most two 5-classes, because at most two maximum stable sets are disjoint. Profile arithmetic and bounds on the 4-set layer, some of them conditioned on the choice of 5-class, leave 16 candidate profiles, seven with one 5-class and nine with two. For each possible choice of 5-classes we then bound how many 4-sets, and then 3-sets and 2-sets, fit into the remaining vertices, and every candidate asks for more than its choice allows.

4.1 The graph, a coloring of sum 8,277, and the maximum stable sets

DSJC250.9 has 250 vertices and 27,897 edges. Appendix B lists a coloring with 73 classes: every vertex receives one label, the labels $1, \dots, 73$ are all used, and no edge joins two vertices with the

Table 2: Reduction of the improving profiles of DSJC250.9. The last row applies, for each possible choice of 5-classes, the residual 4-set bounds of Table 3.

stage	total	$h_5 = 0$	$h_5 = 1$	$h_5 = 2$
profiles with $\Phi(h) \leq 8,276$	6,668	1,741	2,208	2,719
$g_4 \leq 39$; if $h_5 = 2$, $g_4 \leq 36$ (S1, S2)	45	0	36	9
residual 4-set caps for each top packing	16	0	7	9

same label. Its profiles are

$$h = (2, 4, 29, 37, 1), \quad g = (73, 71, 67, 38, 1), \quad (8)$$

so by Proposition 2.1 its sum is

$$\Phi(h) = T(73) + T(71) + T(67) + T(38) + T(1) = 2701 + 2556 + 2278 + 741 + 1 = 8277. \quad (9)$$

Hence $\Sigma(\text{DSJC250.9}) \leq 8,277$, and we take $B = 8,276$.

Enumerating the stable sets in increasing vertex order gives

$$(|S_1|, \dots, |S_6|) = (250, 3,228, 2,869, 205, 3, 0).$$

There is no stable 6-set, so $\alpha = 5$, and there are exactly three stable 5-sets:

$$X_1 = \{3, 69, 88, 154, 223\}, \quad X_2 = \{4, 51, 121, 189, 247\}, \quad X_3 = \{67, 119, 166, 189, 247\}.$$

X_1 is disjoint from X_2 and from X_3 , while $X_2 \cap X_3 = \{189, 247\}$. The top packing of any coloring is therefore one of $\emptyset$, $\{X_1\}$, $\{X_2\}$, $\{X_3\}$, $\{X_1, X_2\}$ and $\{X_1, X_3\}$, and at most two 5-classes can occur; two is the cap of Gondran et al. [7] for this graph. The coloring above has the single 5-class X_2 .

4.2 Profiles that could improve on 8,277

By Corollary 2.2 and Proposition 2.3, a coloring of sum at most 8,276 has an exact-size profile $h = (h_1, \dots, h_5)$ of nonnegative integers with

$$\sum_{s=1}^5 s h_s = 250, \quad h_5 \leq 2, \quad \Phi(h) \leq 8,276. \quad (10)$$

Listing the integer solutions is a finite calculation: there are 6,668 of them. Table 2 shows how they are distributed over h_5 and how the next two steps reduce them.

4.3 The 4-set layer

Let $g_4 = h_4 + h_5$ be the number of classes of size at least four. The certificate S1 described after Lemma 3.3 gives $g_4 \leq 39$ for every coloring, and Appendix B lists 39 pairwise-disjoint stable 4-sets, so 39 is the exact maximum. A second certificate, S2, uses a multiplier for the fixed count $h_5 = 2$; with $D = 2$ and right-hand side 73 it gives $2g_4 \leq 73$, so $g_4 \leq 36$ when there are two 5-classes. Both certificates are checked against every stable 4- and 5-set, and in both the minimum slack is zero.

These two bounds leave 45 of the 6,668 profiles. None has $h_5 = 0$. With $g_5 = 0$ and $g_4 \leq 39$, the cost is least when the first three Ferrers coordinates are balanced, at $g = (71, 70, 70, 39, 0)$, where $\Phi = 8,306 > 8,276$; so a profile with $h_5 = 0$ would need $g_4 \geq 40$. Of the 36 survivors with

Table 3: Packing capacities of DSJC250.9. An entry $= k$ is attained by an explicit packing; $\leq k$ is an upper bound. The 3-set column assumes the maximum number of residual 4-sets; the 2-set column assumes the 4- and 3-set counts required by H1-C (for $\{X_2\}$) and H2-F (for $\{X_1, X_2\}$).

case	4-sets	3-sets	2-sets	certificates
<i>Global bounds on g_4, the number of classes of size at least four</i>				
all colorings	$g_4 = 39$			S1; Appendix B
colorings with $h_5 = 2$	$g_4 \leq 36$			S2
<i>Residual bounds for each top packing</i>				
$\{X_1\}$	≤ 36			S3
$\{X_2\}$	$= 37$	$= 29$	$= 4$	S4, S7, S10
$\{X_3\}$	$= 37$	$= 28$		S5, S8
$\{X_1, X_2\}$	$= 34$	$= 31$	$= 4$	S2, S9, S11
$\{X_1, X_3\}$	≤ 33			S6

one 5-class, 29 have $(h_4, h_5) = (38, 1)$ and hence $g_4 = 39$, and seven have $(h_4, h_5) = (37, 1)$ and $g_4 = 38$. The nine with two 5-classes all have $(h_4, h_5) = (34, 2)$.

Conditioning first pays off at this layer. [Table 3](#) lists what fits below each top packing. If X_1 is the only 5-class, at most 36 disjoint 4-sets fit in the remaining vertices (S3), so $g_4 \leq 37$. If it is X_2 or X_3 , at most 37 fit (S4, S5), so $g_4 \leq 38$. Without conditioning the bound is $g_4 \leq 39$, and it is attained. None of the 29 profiles with $g_4 = 39$ is therefore realizable under any choice of 5-class, while the seven with $g_4 = 38$ survive under X_2 or X_3 . With the nine two-class profiles, $7 + 9 = 16$ candidates remain; they are listed in [Table A.1](#). Every candidate with one 5-class needs 37 residual 4-sets and every candidate with two needs 34.

4.4 The 3-set and 2-set layers

For the survivors the next question is how many 3-sets fit once the required 4-sets are in place. Under X_2 , take the eligible sizes $\{4, 3\}$, the cap $n_4 \leq 37$ from S4, the score $Z_3 = 251n_4 + n_3$ and the target $37 \cdot 251 + 30 = 9,317$. An admissible tree, S7, shows that every packing scores at most 9,316 $= 37 \cdot 251 + 29$, and an explicit packing with 37 4-sets and 29 3-sets attains this score. So with 37 residual 4-sets at most 29 residual 3-sets fit, and 29 is attained. The same construction gives at most 28 3-sets under X_3 (S8: score at most 9,315 $= 37 \cdot 251 + 28$, attained by (37, 28)) and at most 31 3-sets beside 34 4-sets under $\{X_1, X_2\}$ (S9: score at most 8,565 $= 34 \cdot 251 + 31$, attained by (34, 31)).

For the two candidates that survive this layer we go one step further, with $Z_2 = 251^2 n_4 + 251n_3 + n_2$ and $251^2 = 63,001$. Under X_2 with $n_4 \leq 37$ and $n_3 \leq 29$, the tree S10 bounds Z_2 by 2,338,320, one less than the target $37 \cdot 63,001 + 29 \cdot 251 + 5 = 2,338,321$. So at most four 2-sets fit beside 37 4-sets and 29 3-sets, and the coloring of [Appendix B](#) attains (37, 29, 4). Under $\{X_1, X_2\}$ with $n_4 \leq 34$ and $n_3 \leq 31$, S11 bounds Z_2 by 2,149,819 against the target 2,149,820. So at most four 2-sets fit beside 34 4-sets and 31 3-sets; this is attained by a coloring of sum 8,278 with profile (3, 4, 31, 34, 2).

Every entry marked $=$ in [Table 3](#) is attained by an explicit packing. These are therefore the exact conditional maxima, not merely upper bounds.

4.5 The seven branches

[Table 4](#) lists the complete case analysis, and [Figure 2](#) draws it as a tree.

Candidates with one 5-class. Each needs 37 residual 4-sets. Under X_1 at most 36 fit (branch 1). Under X_3 , 37 4-sets leave room for at most 28 3-sets, while every candidate needs at

Table 4: The seven branches of the DSJC250.9 proof. Each row is an application of [Proposition 3.2](#) to the complete residual universe of the stated top packing.

branch	case	certified consequence	candidates closed
1	one 5-class, X_1	S3: $n_4 \leq 36$	all seven (each needs $n_4 = 37$)
2	one 5-class, X_3	S5, S8: $n_4 = 37 \Rightarrow n_3 \leq 28$	all seven (each needs $n_3 \geq 29$)
3	one 5-class, X_2 , $n_3 \geq 30$	S4, S7: $n_4 = 37 \Rightarrow n_3 \leq 29$	six
4	H1-C under X_2	S10: $(n_4, n_3) = (37, 29) \Rightarrow n_2 \leq 4$	H1-C, which needs $n_2 = 5$
5	two 5-classes, $\{X_1, X_3\}$	S6: $n_4 \leq 33$	all nine (each needs $n_4 = 34$)
6	two 5-classes, $\{X_1, X_2\}$, $n_3 \geq 32$	S2, S9: $n_4 = 34 \Rightarrow n_3 \leq 31$	eight
7	H2-F under $\{X_1, X_2\}$	S11: $(n_4, n_3) = (34, 31) \Rightarrow n_2 \leq 4$	H2-F, which needs $n_2 = 5$

least 29 (branch 2). Under X_2 , 37 4-sets leave room for at most 29 3-sets, which rules out the six candidates that need 30 or more (branch 3). The last candidate is

$$\text{H1-C} = (h_1, h_2, h_3, h_4, h_5) = (0, 5, 29, 37, 1), \quad \Phi = 8,276, \quad (11)$$

which needs X_2 as its 5-class, 37 4-sets, 29 3-sets and five 2-sets; at most four fit (branch 4).

Candidates with two 5-classes. The top packing is $\{X_1, X_2\}$ or $\{X_1, X_3\}$, and each candidate needs 34 residual 4-sets. Under $\{X_1, X_3\}$ at most 33 fit (branch 5). Under $\{X_1, X_2\}$, 34 4-sets leave room for at most 31 3-sets, which rules out the eight candidates that need 32 or more (branch 6). The last candidate is

$$\text{H2-F} = (h_1, h_2, h_3, h_4, h_5) = (1, 5, 31, 34, 2), \quad \Phi = 8,276, \quad (12)$$

which needs 34 4-sets, 31 3-sets and five 2-sets; at most four fit (branch 7).

The last two branches have a transparent reading. Under H1-C the 5-class, the 37 4-sets and the 29 3-sets cover $5 + 148 + 87 = 240$ vertices, and the profile needs the remaining ten to split into five stable pairs, that is, a perfect matching in the complement of the graph they induce. Under H2-F the top classes, 34 4-sets and 31 3-sets cover $10 + 136 + 93 = 239$ vertices, and the remaining eleven would need to form five stable pairs and a singleton. S10 and S11 certify, by dual bounds over the whole residual packing, that at most four pairs are available.

Theorem 4.1. *No proper coloring of DSJC250.9 has sum at most 8,276.*

Proof. By [Corollary 2.2](#) and [Proposition 2.3](#) and the census of [Section 4.1](#), such a coloring has one of the 6,668 profiles of [Section 4.2](#). The certificates S1–S5, applied to every possible top packing, reduce these to the 16 candidates of [Table A.1](#), and the seven branches of [Table 4](#) exclude all of them. $\square$

Proof of Theorem 1.1(i). The coloring (9) gives $\Sigma(\text{DSJC250.9}) \leq 8,277$, and [Theorem 4.1](#) gives $\Sigma(\text{DSJC250.9}) \geq 8,277$. $\square$

4.6 A direct solve of the master

For this graph the integer master of [Section 2.3](#) is small. It has 255 equations and 7,125 variables: 6,555 binary set variables, one for each nonempty stable set, and 570 continuous unit-cell variables. Two archived one-thread runs of HiGHS 1.11.0 each reported objective and bound 8,277 with zero displayed gap, after 1,601 branch-and-bound nodes and 278,786 LP iterations. Both returned

Table 5: Stable sets, top packings and the coloring of sum 29,848 for DSJC500.9.

quantity	value
vertices	500
distinct undirected edges	112,437
$(\mathcal{S}_1 , \dots, \mathcal{S}_6)$	$(500, 12, 313, 19, 901, 2, 428, 23, 0)$
independence number	5
maximum number of disjoint stable 5-sets	15
number of maximum 5-set packings	8
profile $(h_1, \dots, h_5)$ of the coloring	$(3, 4, 10, 96, 15)$
classes / sum of the coloring	128 / 29,848

the same 73-class coloring, which we checked independently to have profile $(2, 4, 29, 37, 1)$ and sum 8,277. The model’s column universe and coefficients were also checked independently.

The runs agree with [Theorem 4.1](#) but do not prove it: the solver works in floating point, and the archive holds no independently replayable record of its lower bound. The argument of this section supplies such a proof, and one whose structure a reader can follow. Eleven certificates in seven branches show that under every choice of 5-classes the remaining vertices lack room for the 4-, 3- or 2-sets an improving profile needs. The archived runs are described further in [Appendix E.4](#).

5 DSJC500.9: how far conditioning goes

For DSJC500.9 the same approach proves the lower bound 29,791 but leaves an interval. The three cases at its lower end show where reasoning about the conditioned relaxation stops.

5.1 The graph, a coloring and the top packings

The census of DSJC500.9 is in [Table 5](#). Its DIMACS file declares 224,874 edges and lists 112,437 distinct undirected edges, a documented doubled-header convention ([Appendix E](#)). There are 23 stable 5-sets and no stable 6-set, so $\alpha = 5$. At most 15 of the 5-sets are pairwise disjoint, and exactly eight families of 15 disjoint 5-sets exist; we number them P1–P8 in the deterministic order used by the checking programs.

A coloring with 128 classes has profile $(3, 4, 10, 96, 15)$ and sum 29,848; every vertex occurs once, each class is stable, and the labels are already in nonincreasing order of class size. Its fifteen 5-classes form P5. Every coloring of smaller sum must also have fifteen 5-classes. If $g_5 = h_5 \leq 14$, the cost is least when the first four Ferrers coordinates are balanced, and while $g_5 \leq 14$ moving a unit from one of them (each at least 121) to g_5 lowers the cost. The minimum is therefore at $g = (122, 122, 121, 121, 14)$, where

$$2T(122) + 2T(121) + T(14) = 29,873 > 29,847.$$

So the top packing of an improving coloring is one of P1–P8. Each covers 75 vertices and leaves 425 for classes of size at most four. Here a cell is a pair of one of these packings and a profile; the same profile under two packings gives two cells.

5.2 Excluding cells

Set $B = 29,847$. Graph-free enumeration gives 230 profiles with $\Phi(h) \leq B$, all with $h_5 = 15$. Residual 4-set certificates give $h_4 \leq 101$ under each of the eight packings, which leaves 160

Table 6: Exclusion of cells for DSJC500.9.

stage	removed	remaining
graph-free profiles with $\Phi(h) \leq 29,847$, all with $h_5 = 15$	–	230 profiles
residual bound $h_4 \leq 101$ under each of P1–P8	–	$8 \times 160 = 1,280$ cells
Farkas certificates	11	1,269
$h_4 \leq 100$ under P2 and P4 (two trees)	74	1,195
conditional trees (17 trees)	30	1,165

Table 7: The three remaining DSJC500.9 cells of value 29,791.

top packing	$(h_1, h_2, h_3, h_4, h_5)$	$\Phi(h)$
P5	(1, 4, 4, 101, 15)	29,791
P5	(4, 1, 5, 101, 15)	29,791
P7	(4, 1, 5, 101, 15)	29,791

profiles per packing, 1,280 cells in all. Farkas certificates exclude the profile $(0, 0, 7, 101, 15)$ under each of the eight packings and three further cells, leaving 1,269. Two further 4-set trees sharpen the bound to $h_4 \leq 100$ under P2 and P4, removing 74 cells.

Seventeen conditional trees then exclude 30 more cells, each through an explicit implication from a cell to a model of [Proposition 3.2](#). The implication checks the top packing, the complete residual universe, the caps and the score that the cell requires. For example, under the caps $h_4 \leq 101$ and $h_3 \leq 6$, a profile with $(h_4, h_3) = (101, 6)$ has score $501h_4 + h_3 = 50,607$, and a tree showing that every compatible residual packing scores less excludes it. Other trees also carry the simultaneous demand for pairs where it is needed. [Table 6](#) summarizes the reduction; 1,165 cells remain. (A one-way splitting organization of these cells, which no exclusion uses, is described in [Appendix C](#).)

5.3 The lower bound

Every cell of value at most 29,790 is excluded, and exactly three cells remain at 29,791 ([Table 7](#)). Profile arithmetic alone does not give this value: the graph-free profiles with $\Phi(h) \leq B$ take every value from 29,766 to 29,847. Those of value 29,789 and 29,790 all have $h_4 \geq 102$ and are removed by the residual bound $h_4 \leq 101$, and among the profiles with $h_4 \leq 101$ no value lies strictly between 29,788 and 29,791.

Theorem 5.1. $29,791 \leq \Sigma(\text{DSJC500.9}) \leq 29,848$.

Proof. The coloring above gives the upper bound. A coloring of smaller sum has one of P1–P8 as its top packing and one of the enumerated profiles, and the certificates of [Section 5.2](#) exclude every such cell of value at most 29,790. $\square$

This proves [Theorem 1.1\(ii\)](#). Of the 1,165 remaining cells, three have value 29,791 and 1,162 have values from 29,792 to 29,847. Excluding the three lowest would raise the lower bound to 29,792. The census decides neither whether 29,791 is attained nor whether the coloring of sum 29,848 is optimal. [Appendix D](#) records a separate certificate about that coloring: every improving coloring must change at least one of its 5- or 4-classes.

5.4 Why the three lowest cells resist

The three endpoint cells need joint residual packings. Under P5 or P7, 101 residual 4-sets must coexist with four 3-sets and four 2-sets, or with five 3-sets and one 2-set. The difficulty lies in the identities and compatibility of several layers at once, not in the number of 4-sets alone.

A continuous model shows that no argument valid for the conditioned relaxation can exclude these cells. Fix the top packing P of a cell, let $R = V \setminus \bigcup_{S \in P} S$, and let $\mathcal{S}_s(R)$ be the family of stable s -sets of $G[R]$. For the cell's counts (h_2, h_3, h_4) consider

$$\begin{aligned} x_S &\geq 0 & (S \in \mathcal{S}_2(R) \cup \mathcal{S}_3(R) \cup \mathcal{S}_4(R)), \\ \sum_{S \ni v} x_S &\leq 1 & (v \in R), \\ \sum_{S \in \mathcal{S}_s(R)} x_S &= h_s & (s = 2, 3, 4). \end{aligned} \tag{13}$$

Given a solution, give each residual vertex the singleton weight $1 - \sum_{S \ni v} x_S$. These weights are nonnegative and sum to $425 - 2h_2 - 3h_3 - 4h_4 = h_1$, so together with the top packing they satisfy the covering equations of the master relaxation conditioned on P . The class totals are integral even though the column weights need not be, so the Ferrers coordinates are those of the cell and the unit-cell objective is exactly $\Phi(h) = 29,791$.

Explicit rational solutions of (13) exist for P5 with $(h_2, h_3, h_4) = (4, 4, 101)$, and for P5 and for P7 with $(h_2, h_3, h_4) = (1, 5, 101)$. Their checks reconstruct the graph and the packings, verify that every column in the support is a stable residual set, and recompute the vertex loads, the class totals and the singleton deficits exactly (Appendix E.3). Hence no inequality valid for this continuous model excludes the three cells. Excluding them requires information of another kind, such as integrality of the residual packing, a disjunction or branching, or a valid inequality for the integer hull. The rational points establish continuous feasibility only; they are not colorings, and they do not rule out stronger relaxations.

6 The master relaxation: DSJC1000.9 and C2000.9

For the two larger graphs $\alpha = 6$, and we use the master relaxation of Section 2.3 rather than integer packings. DSJC1000.9 has only three stable 6-sets, so conditioning on them gives an eight-way case split with one relaxation for each case. For C2000.9 the cap on the number of 6-classes already gives a bound that the unconditioned relaxation attains.

6.1 DSJC1000.9

DSJC1000.9 has 1,000 vertices and 449,449 edges. Complete enumeration gives

$$(|\mathcal{S}_1|, \dots, |\mathcal{S}_6|) = (1,000, 50,051, 167,104, 41,659, 869, 3)$$

and no stable 7-set, so $\alpha = 6$ and there are 260,686 nonempty stable sets. The three stable 6-sets are

$$\begin{aligned} A_1 &= \{52, 54, 175, 319, 501, 704\}, \\ A_2 &= \{76, 145, 184, 672, 793, 805\}, \\ A_3 &= \{129, 150, 181, 321, 456, 762\}, \end{aligned}$$

and they are pairwise disjoint, so the 6-classes of any coloring form one of their eight subsets. Unlike the top packings of DSJC500.9, where the number of 5-classes was forced to its maximum, these cases include every number of 6-classes from zero to three.

The unconditioned relaxation. A rational dual solution of the master relaxation, checked against all 260,686 columns, has value

$$L_{\text{master}} = \frac{102308106541886}{10^9} = 102,308.106541886,$$

so $\Sigma(\text{DSJC1000.9}) \geq 102,309$. This dual is feasible but not optimal: every column inequality holds with slack at least $|S| \cdot 10^{-9}$, so raising every vertex weight by 10^{-9} preserves feasibility and increases the value by 10^{-6} . We have not computed the optimum of the relaxation.

The eight conditioned relaxations. For a mask $b = (b_1, b_2, b_3) \in \{0, 1\}^3$, select A_i exactly when $b_i = 1$, let $k = b_1 + b_2 + b_3$, and put

$$R_b = V \setminus \bigcup_{i: b_i=1} A_i, \quad \mathcal{S}_b = \{S \subseteq R_b : S \text{ stable}, 1 \leq |S| \leq 5\}.$$

An unselected A_i is excluded as a whole class, but its smaller stable subsets remain eligible, so $\mathcal{S}_b$ is the complete residual universe for this choice of 6-classes. The leftmost bit refers to A_1 ; for example, mask 101 selects A_1 and A_3 and excludes A_2 as a class. With $N_t = \lfloor |R_b|/t \rfloor$, the conditioned relaxation is

$$\begin{aligned} & \text{minimize} && 6T(k) + \sum_{t=1}^5 \sum_{j=1}^{N_t} (k+j) u_{t,j} \\ & \text{subject to} && \sum_{S \in \mathcal{S}_b: v \in S} x_S = 1 && (v \in R_b), \\ & && r_t = \sum_{S \in \mathcal{S}_b: |S| \geq t} x_S = \sum_{j=1}^{N_t} u_{t,j} && (1 \leq t \leq 5), \\ & && x_S \geq 0, && 0 \leq u_{t,j} \leq 1. \end{aligned} \tag{14}$$

For an integral cover, r_t counts the residual classes of size at least t , the full Ferrers coordinates are $q_t = k + r_t$ for $t \leq 5$ and $q_6 = k$, and the minimized objective over the unit cells is

$$6T(k) + \sum_{t=1}^5 (kr_t + T(r_t)) = \sum_{t=1}^6 T(q_t).$$

The shift by k accounts for the labels already taken by the k selected 6-classes. For fractional x the coordinates r_t need not be integers, and the model keeps the piecewise-linear unit-cell cost rather than substituting $T(r_t)$.

Scenario 111 receives one additional inequality. Its 982 residual vertices contain exactly 750 stable 5-sets, and nonnegative rational weights w_v satisfy

$$\sum_{v \in S} w_v \geq 1 \quad (S \in \mathcal{S}_{111}, |S| = 5), \quad \sum_{v \in R_{111}} w_v = \frac{134549891193}{10^9} < 135.$$

Summing over a disjoint family of these 5-sets shows that its integer cardinality is strictly below 135, hence at most 134. Every integral coloring in scenario 111 therefore has at most 134 residual 5-classes. Thus $r_5 \leq 134$ is an integer-valid strengthening imposed on (14) for $b = 111$ only. For a fractional packing the weight vector alone gives only $r_5 \leq 134.549891193$. The cap concerns residual classes; the corresponding full coordinate is $q_5 = 3 + r_5$. Both the strict threshold 135 and the residual universe after all three selections are essential to the inference.

Table 8: The eight cases for DSJC1000.9. In each, the listed 6-sets are selected and the others among A_1, A_2, A_3 are excluded as classes. Every decimal is an exact rational with denominator 10^9 ; only mask 111 uses the cap $r_5 \leq 134$.

mask	selected 6-sets	L_b	$\lceil L_b \rceil$
000	none	102,633.344579608	102,634
001	A_3	102,513.319748705	102,514
010	A_2	102,557.583397861	102,558
011	A_2, A_3	102,432.711355035	102,433
100	A_1	102,503.510602511	102,504
101	A_1, A_3	102,395.380611244	102,396
110	A_1, A_2	102,443.215571129	102,444
111	A_1, A_2, A_3	102,343.581115274	102,344

Dual certificates. Each scenario is certified by rational vertex prices π_v ($v \in R_b$), slopes β_t and a multiplier $\delta \geq 0$ for the cap, with $\delta = 0$ except in scenario 111, satisfying

$$\sum_{v \in S} \pi_v \leq \sum_{t=1}^{|S|} \beta_t + \delta \mathbf{1}_{\{|S|=5\}} \quad (S \in \mathcal{S}_b). \quad (15)$$

Put $\Gamma_k(\beta) = \sum_{j \geq 1} \max\{0, \beta - (k + j)\}$, a finite sum. The certificates have $\beta_t \leq k + N_t + 1$, so the positive terms have $j \leq N_t$. Each unit cell satisfies $(k + j)u \geq \beta u - \max\{0, \beta - (k + j)\}$ for $0 \leq u \leq 1$, so the objective of (14) is at least $6T(k) + \sum_t \beta_t r_t - \sum_t \Gamma_k(\beta_t)$. By (15) and the exact cover, $\sum_t \beta_t r_t = \sum_S x_S \sum_{t \leq |S|} \beta_t \geq \sum_v \pi_v - \delta r_5$, and $r_5 \leq 134$ in scenario 111. Hence

$$L_b = 6T(k) + \sum_{v \in R_b} \pi_v - \sum_{t=1}^5 \Gamma_k(\beta_t) - 134\delta \quad (16)$$

is a lower bound for scenario b , the last term being present only for $b = 111$. The certificates use the denominator 10^9 throughout; their inequalities are checked over all eligible residual columns and their values are recomputed in integer arithmetic.

Every coloring belongs to one row of Table 8, so

$$\Sigma(\text{DSJC1000.9}) \geq \left\lceil \min_{b \in \{0,1\}^3} L_b \right\rceil = \left\lceil \frac{102343581115274}{10^9} \right\rceil = 102,344.$$

Conditioning adds 35 to the bound 102,309 from the unconditioned dual. Because that dual is not optimal, the 35 measures the gain over this dual, not over the optimum of the unconditioned relaxation.

A coloring with 225 classes and profile $(h_1, \dots, h_6) = (2, 4, 17, 74, 125, 3)$, checked class by class and edge by edge, has sum 103,256. Hence

$$102,344 \leq \Sigma(\text{DSJC1000.9}) \leq 103,256, \quad (17)$$

which proves Theorem 1.1(iii); the gap is 912.

6.2 C2000.9

C2000.9 has 2,000 vertices and 1,799,532 distinct undirected edges. Complete enumeration gives

$$(|\mathcal{S}_1|, \dots, |\mathcal{S}_6|) = (2,000, 199,468, 1,322,912, 656,504, 26,224, 90)$$

and no stable 7-set, so $\alpha = 6$. Among the 90 stable 6-sets the certificate exhibits 52 that are pairwise disjoint, and partitions all 90 into 52 cliques of their intersection graph. A disjoint family contains at most one member of each clique, so the maximum is exactly 52, and every coloring has $h_6 \leq 52$. (Any 52-vertex set meeting every 6-set, such as the set Z used below, yields such a partition: group the 6-sets according to a vertex of Z that they contain.)

The profile bound. With $\alpha = 6$ and at most 52 classes of size six, the profile relaxation of Gondran et al. [7, Sections 2.2–2.3] gives the lower bound. For fixed $g_6 = m \leq 52$ the first five Ferrers coordinates sum to $2000 - m$, and by convexity of T their contribution is least when they differ by at most one. Increasing m towards 52 lowers the total, since it moves a unit from a coordinate of size at least 389 to one of size at most 52. The unique minimizer is

$$g = (390, 390, 390, 389, 389, 52), \quad h = (0, 0, 1, 0, 337, 52),$$

with cost

$$3T(390) + 2T(389) + T(52) = 381,823.$$

Our contribution for this graph is the certified value 52 of the cap and the proof, below, that the master relaxation gives nothing more.

The same cap bounds the chromatic number. A coloring with k classes has

$$2000 \leq 6h_6 + 5(k - h_6) = 5k + h_6 \leq 5k + 52, \quad \text{so} \quad k \geq \left\lceil \frac{1948}{5} \right\rceil = 390,$$

which is the lower bound in the range $390 \leq \chi(\text{C2000.9}) \leq 400$ reported by Zheng et al. [24, Table EC.2]. The minimizer above has $g_1 = 390$ and so meets this bound on the number of classes.

The coloring. The upper bound comes from a coloring obtained from one of sum 382,384 by keeping its 310 classes of size five and 52 of size six and recoloring the remaining 138 vertices. The result has 402 classes and profile $(h_1, \dots, h_6) = (1, 4, 11, 24, 310, 52)$; checking the partition, every edge and the objective gives sum 382,379. Hence

$$381,823 \leq \Sigma(\text{C2000.9}) \leq 382,379, \tag{18}$$

which proves [Theorem 1.1\(iv\)](#).

The master relaxation. The relaxation (4) attains the lower endpoint. Its dual has the same form as (15), with $k = 0$, six layers and no cap, so its value is $\sum_v \pi_v - \sum_{t=1}^6 \Gamma_0(\beta_t)$. A 52-vertex set Z meets every 6-set. Put $\pi_v = 52$ for $v \in Z$ and $\pi_v = 390$ otherwise, with slopes $(390, 390, 390, 390, 390, 52)$. For a stable set of size at most five the column inequality follows from $\pi_v \leq 390$. A 6-set meets Z , so

$$\sum_{v \in S} \pi_v \leq 5 \cdot 390 + 52.$$

The affine functions $390q - T(389)$ and $52q - T(51)$ support the unit-cell cost on the first five layers and on the sixth, and weak duality gives the value

$$1948 \cdot 390 + 52 \cdot 52 - 5T(389) - T(51) = 381,823.$$

Two exact rational primal solutions have threshold coordinates

$$\begin{aligned} q &= (1948/5, 1948/5, 1948/5, 1948/5, 1948/5, 52), \\ q &= (390, 390, 1948/5, 1946/5, 1946/5, 52). \end{aligned}$$

Their stable-set weights cover every vertex exactly once, respect the floor caps, and have piecewise-linear cost 381,823. The first proves the optimum of the relaxation, and the second proves the same value under the additional restriction $q_1 \geq 390$. These solutions are fractional and are not colorings. Equality in the integer bound would require exactly 52 classes of size six, 337 of size five and one of size three, and whether such a partition exists is open.

7 How the results are checked

Every lower bound in this paper is an implication from finitely many exact inequalities to a statement about all colorings of a specific graph. It is only as good as the correspondence between the two, so the checking programs in the companion release ([Appendix E](#)) work from the graph files. They reconstruct the stable-set families, including the absence of larger stable sets that fixes α . For each conditional model they rebuild the top classes, the residual vertices, the eligible sizes, the count conditions, the scores and the target, and regenerate the complete residual universe in canonical order. They verify every dual inequality of [Lemma 3.3](#) or (15) column by column in integer arithmetic, check that every tree is admissible in the sense of [Section 3.4](#), and check every coloring edge by edge and recompute its sum. A profile exclusion then follows from [Proposition 3.2](#), or, for DSJC1000.9, from the conditioned relaxation (14).

This is the division of labour of safe column generation [8]: inexact computations propose dual solutions, and exact computations certify them. Here the certification is a direct check against enumerated columns, which the density of the graphs makes possible. The resulting certificates are specific to this setting. General formats for integer programming [2], including certificates constructed from black-box ILP solvers [19], and proof logging for pseudo-Boolean optimization [10] certify solver runs on arbitrary models. Dold et al. [4] obtain end-to-end formally certified results for graph coloring by combining proof logging with a formally verified checker. Our checkers are ordinary programs using exact integer arithmetic, and the correspondence between each theorem and the certificates it uses is set out in [Tables 4](#) and [A.2](#) and in the text.

What must be trusted is therefore the checking code; the exact integer arithmetic of its runtime; the SHA-256 and decompression implementations that bind files to the identities of [Table E.1](#); and the human correspondence between theorem statements and certificate scopes. The programs that generated the certificates and the floating-point solvers are not part of this base. Available generation records and timings are included in the companion release. The archived HiGHS runs of [Section 4.6](#) are checked statically: the checker validates the model and the returned colorings, while the solver’s reported optimality keeps the status described there.

8 Discussion

The profile representation separates the cost of a coloring from the compatibility of its classes. Conditional envelopes reconnect the two by preserving the selected largest sets and certifying what remains possible below them. On DSJC250.9 envelopes at three layers suffice: under every choice of 5-classes there is too little room for the 4-, 3- or 2-sets that an improving profile would need. On DSJC500.9 the same construction leaves 1,165 cells, and the rational points at the three lowest show that the conditioned continuous model cannot finish the exclusion by itself. On the two larger graphs we used the master relaxation instead. On DSJC1000.9 conditioning on eight choices adds 35 to a feasible dual bound, and on C2000.9 the global cap on 6-classes already meets the optimum of the unconditioned relaxation.

Conditioning is a disjunction over the possible top layers, much as branching in branch-and-price for coloring [3, 15] splits the solution space. Here the disjunction is taken over the top Ferrers layer, where the three DSJC graphs offer few relevant possibilities. Bounds on the integer packings below this layer exclude every improving candidate for DSJC250.9; for DSJC500.9

they leave the unresolved cells described in [Section 5](#).

The conditioned arguments need three things: an independence number small enough that the stable sets of the top sizes can be listed, few relevant top packings to condition on, and residual families small enough for exact duals and branch trees. The three DSJC graphs studied here have all three; C2000.9 uses the global cap and unconditioned master without enumerating top packings. For other graphs the size of the column families and, in conditioned arguments, the number of top cases would have to be controlled anew, and these four cases give no general complexity guarantee.

Three concrete questions remain. Excluding the three cells of [Table 7](#) would raise the DSJC500.9 bound to 29,792. For DSJC1000.9, the multiplier of the cap in scenario 111 is $\delta = 65.729495535$; a proof that $r_5 \leq 133$ there, that is, that at most 133 disjoint stable 5-sets fit in the 982 vertices outside $A_1 \cup A_2 \cup A_3$, would raise L_{111} to 102,409.310610809, making scenario 101 the minimum and the bound 102,396. No certificate for that cap is available. For C2000.9, the lower bound 381,823 is attained if and only if the vertices can be partitioned into 52 stable 6-sets, 337 stable 5-sets and one stable 3-set.

Use of generative AI

Generative AI tools assisted with mathematical exploration, software development, and manuscript drafting and revision. The author reviewed the arguments and text and takes responsibility for the paper. The computational claims are supported by the accompanying certificates and verification programs.

A Proof data for DSJC250.9

A.1 The sixteen candidates

Table A.1: The DSJC250.9 profiles that survive the 4-set layer at the threshold 8,276 ([Section 4.3](#)): seven with one 5-class, then nine with two.

h_1	h_2	h_3	h_4	h_5	classes	g	$\Phi(h)$
0	5	29	37	1	72	(72, 72, 67, 38, 1)	8276
3	2	30	37	1	73	(73, 70, 68, 38, 1)	8274
1	3	30	37	1	72	(72, 71, 68, 38, 1)	8272
4	0	31	37	1	73	(73, 69, 69, 38, 1)	8273
2	1	31	37	1	72	(72, 70, 69, 38, 1)	8270
0	2	31	37	1	71	(71, 71, 69, 38, 1)	8269
1	0	32	37	1	71	(71, 70, 70, 38, 1)	8268
1	5	31	34	2	73	(73, 72, 67, 36, 2)	8276
4	2	32	34	2	74	(74, 70, 68, 36, 2)	8275
2	3	32	34	2	73	(73, 71, 68, 36, 2)	8272
0	4	32	34	2	72	(72, 72, 68, 36, 2)	8271
5	0	33	34	2	74	(74, 69, 69, 36, 2)	8274
3	1	33	34	2	73	(73, 70, 69, 36, 2)	8270
1	2	33	34	2	72	(72, 71, 69, 36, 2)	8268
2	0	34	34	2	72	(72, 70, 70, 36, 2)	8267
0	1	34	34	2	71	(71, 71, 70, 36, 2)	8266

A.2 The eleven certificates

Table A.2 records the model and conclusion of each certificate used in Section 4. Each residual family is the complete family of stable sets of the indicated sizes disjoint from the top packing. S1 and S2 are global bounds with no target; the other rows state their target in the model column. Every tree closes with integer-dual leaves. The file stems in the second column identify the descriptors in the companion release.

Table A.2: Certificates S1–S11 for DSJC250.9.

ID	file stem; top packing	model, caps, target	proof	consequence
S1	ge4_global; none	sizes 4, 5; score 1	rational $D = 26$	$g_4 \leq 39$
S2	ge4_h5_eq_2; $h_5 = 2$	sizes 4, 5; score 1; equality $h_5 = 2$	rational $D = 2$	$g_4 \leq 36$
S3	quad_x1; $\{X_1\}$	residual 4-sets; score n_4 ; target 37	rational $D = 3$	$n_4 \leq 36$
S4	quad_x2; $\{X_2\}$	residual 4-sets; score n_4 ; target 38	tree	$n_4 \leq 37$
S5	quad_x3; $\{X_3\}$	residual 4-sets; score n_4 ; target 38	rational $D = 25$	$n_4 \leq 37$
S6	quad_x1x3; $\{X_1, X_3\}$	residual 4-sets; score n_4 ; target 34	rational $D = 2$	$n_4 \leq 33$
S7	env_x2; $\{X_2\}$	residual 4, 3-sets; score $251n_4 + n_3$; cap $n_4 \leq 37$; target 9317	tree	$n_4 = 37 \Rightarrow n_3 \leq 29$
S8	env_x3; $\{X_3\}$	residual 4, 3-sets; score $251n_4 + n_3$; cap $n_4 \leq 37$; target 9316	tree	$n_4 = 37 \Rightarrow n_3 \leq 28$
S9	env_x1x2; $\{X_1, X_2\}$	residual 4, 3-sets; score $251n_4 + n_3$; cap $n_4 \leq 34$; target 8566	tree	$n_4 = 34 \Rightarrow n_3 \leq 31$
S10	pair_H1C; $\{X_2\}$	residual 4, 3, 2-sets; score $63001n_4 + 251n_3 + n_2$; caps $n_4 \leq 37$, $n_3 \leq 29$; target 2338321	tree	$(n_4, n_3) = (37, 29) \Rightarrow n_2 \leq 4$
S11	pair_H2F; $\{X_1, X_2\}$	residual 4, 3, 2-sets; score $63001n_4 + 251n_3 + n_2$; caps $n_4 \leq 34$, $n_3 \leq 31$; target 2149820	tree	$(n_4, n_3) = (34, 31) \Rightarrow n_2 \leq 4$

A.3 Tree sizes and single-dual values

Table A.3: Sizes of the DSJC250.9 branch trees. Columns are the eligible residual stable sets at the root. Every leaf is closed by an integer dual certificate (Corollary 3.4).

tree	columns	nodes	branches	leaves
quad_x2 (S4)	178	75	37	38
env_x3 (S8)	2,880	155	77	78
env_x1x2 (S9)	2,550	503	251	252
env_x2 (S7)	2,858	1,661	830	831
pair_H2F (S11)	5,500	993	496	497
pair_H1C (S10)	5,953	1,459	729	730

The two global certificates have

$$(D, \text{rhs}, \lfloor \text{rhs}/D \rfloor) = (26, 1039, 39) \quad (\text{S1}), \quad (2, 73, 36) \quad (\text{S2, with } h_5 = 2).$$

The three single residual duals are

certificate	D	rhs	strict target
quad_x1 (S3)	3	109	37
quad_x3 (S5)	25	937	38
quad_x1x3 (S6)	2	67	34

and certify residual bounds of 36, 37 and 33 4-sets. They are checked against 160, 183 and 138 residual columns respectively, and in each case the minimum slack is zero.

B Explicit DSJC250.9 witnesses

B.1 Thirty-nine disjoint stable 4-sets

The following 39 sets are pairwise disjoint and stable in DSJC250.9. They show that the bound $g_4 \leq 39$ of S1 is attained. Their stability and disjointness were rechecked separately from the DIMACS edge set.

stable 4-sets		
{2, 40, 75, 148}	{3, 84, 144, 154}	{4, 121, 189, 247}
{5, 58, 158, 216}	{6, 29, 138, 184}	{7, 95, 109, 244}
{9, 164, 211, 235}	{10, 15, 46, 130}	{11, 103, 202, 228}
{12, 88, 163, 179}	{13, 19, 76, 162}	{16, 145, 182, 220}
{17, 45, 50, 233}	{18, 42, 104, 175}	{23, 124, 168, 207}
{26, 57, 96, 215}	{27, 78, 91, 174}	{28, 167, 194, 218}
{31, 44, 225, 237}	{33, 69, 120, 126}	{34, 56, 219, 236}
{35, 106, 113, 234}	{36, 90, 223, 245}	{39, 52, 70, 136}
{51, 98, 152, 176}	{59, 94, 157, 178}	{63, 139, 209, 227}
{64, 97, 112, 222}	{65, 67, 140, 170}	{71, 87, 180, 200}
{72, 110, 119, 192}	{81, 116, 134, 196}	{83, 183, 187, 226}
{99, 105, 118, 246}	{114, 177, 214, 229}	{143, 188, 198, 243}
{156, 171, 206, 210}	{181, 208, 221, 230}	{191, 205, 232, 250}

B.2 A coloring of sum 8,277

The classes in Table B.1 are listed with their labels. A direct edge scan verifies that every class is stable, and the classes partition $\{1, \dots, 250\}$. Their sizes are nonincreasing, with exact-size profile $(h_1, \dots, h_5) = (2, 4, 29, 37, 1)$; class 1 is X_2 .

Table B.1: A 73-class coloring of DSJC250.9 with sum 8,277.

label: vertex set		
1: {4, 51, 121, 189, 247}	2: {69, 92, 126, 209}	3: {81, 116, 134, 196}
4: {5, 146, 152, 176}	5: {156, 171, 206, 210}	6: {11, 103, 202, 228}
7: {10, 15, 46, 130}	8: {35, 44, 113, 234}	9: {63, 87, 194, 237}
10: {151, 158, 182, 186}	11: {27, 78, 91, 174}	12: {102, 143, 188, 243}
13: {36, 90, 223, 245}	14: {64, 97, 112, 222}	15: {111, 181, 221, 230}
16: {59, 94, 157, 178}	17: {26, 57, 96, 215}	18: {117, 220, 225, 244}
19: {29, 65, 140, 170}	20: {67, 101, 125, 198}	21: {12, 88, 163, 179}
22: {106, 160, 184, 226}	23: {33, 77, 83, 195}	24: {48, 180, 200, 217}
25: {17, 45, 50, 233}	26: {72, 110, 119, 192}	27: {13, 19, 76, 162}
28: {9, 164, 211, 235}	29: {114, 177, 214, 229}	30: {2, 40, 75, 148}
31: {34, 56, 219, 236}	32: {84, 115, 144, 154}	33: {18, 42, 104, 175}
34: {99, 105, 118, 246}	35: {23, 124, 168, 207}	36: {191, 205, 232, 250}
37: {28, 167, 218, 238}	38: {1, 6, 39, 231}	39: {3, 95, 108}
40: {123, 166, 187}	41: {31, 190, 197}	42: {73, 93, 185}
43: {52, 70, 136}	44: {14, 98, 131}	45: {68, 85, 155}
46: {22, 122, 133}	47: {74, 129, 153}	48: {47, 227, 249}
49: {7, 132, 248}	50: {24, 204, 208}	51: {60, 82, 213}
52: {128, 149, 239}	53: {86, 150, 161}	54: {25, 89, 212}
55: {137, 142, 203}	56: {32, 138, 141}	57: {120, 193, 242}
58: {53, 61, 240}	59: {30, 55, 139}	60: {66, 183, 224}
61: {16, 38, 199}	62: {79, 147, 165}	63: {41, 80, 201}
64: {20, 127, 241}	65: {58, 169, 173}	66: {43, 109, 145}
67: {21, 54, 159}	68: {37, 135}	69: {71, 216}
70: {8, 49}	71: {107, 172}	72: {100}
73: {62}		

C Class splitting: an organization of the candidates

Splitting a stable class into smaller classes preserves feasibility and increases the cost. This gives one-way implications between profiles, which organize a finite family of candidates. No proof in this paper depends on these implications: the exclusions established in [Sections 4](#) and [5](#) are certified directly. We record the organization because it shows how the candidates are related, and because the companion release checks it.

Proposition C.1 (One-way splitting). *Let $s = r_1 + \dots + r_m$ with $1 \leq r_i < s$. If a profile h with $h_s > 0$ is feasible, so is the profile obtained by replacing one class of size s by classes of sizes $r_1, \dots, r_m$. Hence, if the latter profile is infeasible, so is h .*

Proof. Partition a stable class of size s into subsets of the prescribed sizes; subsets of a stable set are stable. The second statement is the contrapositive. $\square$

The converse fails in general: merging classes need not preserve feasibility. For DSJC250.9 two moves suffice,

$$\begin{aligned}\sigma_2(h_1, h_2, h_3, h_4, h_5) &= (h_1 + 2, h_2 - 1, h_3, h_4, h_5) & (h_2 > 0), \\ \sigma_3(h_1, h_2, h_3, h_4, h_5) &= (h_1 + 1, h_2 + 1, h_3 - 1, h_4, h_5) & (h_3 > 0),\end{aligned}$$

which split a pair into two singletons and a triple into a pair and a singleton.

Lemma C.2 (Split increments). $\Phi(\sigma_2(h)) - \Phi(h) = h_1 + 1$ and $\Phi(\sigma_3(h)) - \Phi(h) = h_1 + h_2 + 1$.

Proof. σ_2 changes $g_1 \mapsto g_1 + 1$ and $g_2 \mapsto g_2 - 1$, so $\Delta\Phi = (g_1 + 1) - g_2 = h_1 + 1$. σ_3 changes $g_1 \mapsto g_1 + 1$ and $g_3 \mapsto g_3 - 1$, so $\Delta\Phi = (g_1 + 1) - g_3 = h_1 + h_2 + 1$. $\square$

A split can transfer infeasibility within a threshold family $\Phi \leq B$ only if its result stays in the family, and every intermediate profile of a path must do so too. In the graph of [Figure 1](#), splitting the pair $\{2, 4\}$ changes $h = (0, 1, 1)$ to $(2, 0, 1)$ and the cost from 7 to 8, a rise of $h_1 + 1 = 1$; at $B = 7$ the result leaves the family, so this split cannot be used there.

Table C.1: The six splitting-maximal DSJC250.9 candidates.

label	h	g	$\Phi(h)$
H1-A	(4, 0, 31, 37, 1)	(73, 69, 69, 38, 1)	8273
H1-B	(3, 2, 30, 37, 1)	(73, 70, 68, 38, 1)	8274
H1-C	(0, 5, 29, 37, 1)	(72, 72, 67, 38, 1)	8276
H2-D	(5, 0, 33, 34, 2)	(74, 69, 69, 36, 2)	8274
H2-E	(4, 2, 32, 34, 2)	(74, 70, 68, 36, 2)	8275
H2-F	(1, 5, 31, 34, 2)	(73, 72, 67, 36, 2)	8276

For DSJC250.9, the sixteen candidates of [Table A.1](#) reach the six maximal profiles of [Table C.1](#) along the paths of [Table C.2](#), each of which stays at or below 8,276. Excluding the six maximal profiles would therefore exclude all sixteen. The proof in [Section 4.5](#) does not need this: the envelope bounds close every candidate directly, maximal or not.

Table C.2: Splitting paths from each DSJC250.9 candidate to a maximal profile.

candidate	maximal profile	moves
(0, 5, 29, 37, 1)	H1-C	identity
(3, 2, 30, 37, 1)	H1-B	identity

Table C.2: Splitting paths for DSJC250.9 (continued).

candidate	maximal profile	moves
(1, 3, 30, 37, 1)	H1-B	σ_2
(4, 0, 31, 37, 1)	H1-A	identity
(2, 1, 31, 37, 1)	H1-A	σ_2
(0, 2, 31, 37, 1)	H1-A	$\sigma_2 \rightarrow \sigma_2$
(1, 0, 32, 37, 1)	H1-A	$\sigma_3 \rightarrow \sigma_2$
(1, 5, 31, 34, 2)	H2-F	identity
(4, 2, 32, 34, 2)	H2-E	identity
(2, 3, 32, 34, 2)	H2-E	σ_2
(0, 4, 32, 34, 2)	H2-F	σ_3
(5, 0, 33, 34, 2)	H2-D	identity
(3, 1, 33, 34, 2)	H2-D	σ_2
(1, 2, 33, 34, 2)	H2-D	$\sigma_2 \rightarrow \sigma_2$
(2, 0, 34, 34, 2)	H2-D	$\sigma_3 \rightarrow \sigma_2$
(0, 1, 34, 34, 2)	H2-D	$\sigma_2 \rightarrow \sigma_3 \rightarrow \sigma_2$

For DSJC500.9, the 1,195 cells that remain after the bounds $h_4 \leq 100$ under P2 and P4 are organized in the same way, with the splits

$$2 \rightarrow 1 + 1, \quad 3 \rightarrow 2 + 1 \text{ or } 1 + 1 + 1, \quad 4 \rightarrow 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1 \text{ or } 1 + 1 + 1 + 1.$$

From every cell an explicitly checked path reaches one of 178 splitting-maximal targets. Every step keeps the top packing and stays within the admissible family and the threshold. By [Proposition C.1](#) feasibility propagates along these paths, and the converse is not used. The 30 exclusions of [Table 6](#) are exactly the cells justified by the conditional certificate mappings, not by splitting.

D The neighbourhood of the DSJC500.9 coloring

This appendix records a certificate about the coloring of sum 29,848 of [Section 5](#). It does not change the bounds, but it shows that an improvement cannot be found by recoloring only the vertices outside that coloring's classes of sizes five and four.

For a graph H on a vertex set U , the Tutte–Berge formula [\[14\]](#) gives the matching number

$$\nu(H) = \min_{W \subseteq U} \frac{|U| + |W| - \text{odd}(H - W)}{2}, \quad (19)$$

where $\text{odd}(H - W)$ is the number of odd components of $H - W$. A matching and a set W attaining the same value therefore certify $\nu(H)$. Disjoint stable pairs of $G[U]$ are exactly a matching of the complement $H = \overline{G[U]}$.

Fix all 15 classes of size five and all 96 classes of size four of the coloring. The remaining 41 vertices contain no stable 4-set, and at most ten pairwise-disjoint stable 3-sets; there are 12 families of ten. After each of them, the maximum number of disjoint stable pairs among the remaining vertices is at most four, as explicit matchings and Tutte–Berge barriers certify. Of the nine profiles below 29,848 that keep these classes, eight need at least 11 classes of size three, and the ninth needs five pairs beside ten classes of size three. Hence every improving coloring changes at least one of the 5- or 4-classes of this coloring. The statement concerns the identities of those 111 classes; the endpoint cells of [Table 7](#), which have 101 classes of size four, are not constrained by it.

E Files, identities and replay

E.1 Graph files and identities

The graphs are the DIMACS files obtained and checked by the companion release’s fetch script; its graph-provenance document records the CMU COLOR03/COLOR04 sources. A graph file is accepted only with one consistent header, one-based vertex identifiers, no loops and no repeated undirected edges. Declared and parsed edge counts agree except for DSJC500.9, whose header declares 224,874 edges for 112,437 distinct undirected edges, a documented doubled-header convention. Table E.1 lists the SHA-256 identities of the raw and canonical graph files and of the colorings that give the upper bounds; canonical graph hashes are recomputed by the instance checkers. The hashes bind the statements to particular files; checking the mathematics is a separate step.

Table E.1: SHA-256 identities of the graph files and of the colorings giving the upper bounds.

graph	object	SHA-256
DSJC250.9	graph, raw	1b90f811e4d44f8937075865790b845e986b08b9da53caea57b2adf1c57383e7
	graph, canonical	52be422c0065411e126008bf4ee8ee2ba21320bde0a0e94d5ab308c2e34f53a8
	coloring	50665c00da2f606a77b6d83aad8301e3a0ef19819c0da097c64f9d8a8b41ea42
DSJC500.9	graph, raw	95276841dffcd7c04de2fc1ae8e43b6c7b84173b4a9646b1cc6b392a3bd7b39
	graph, canonical	69e7453e2a389197f67fe0e02cf661bc71132b4ea457a434b4726a4b7489b618
	coloring	3f948845ad75ea1ed742916229e6060cab99d6839b7a109af59825e496804760
DSJC1000.9	graph, raw	b73613d4ec2eb988ff124258ebff728a2c08e603c778c7547a15f320a79f61c6
	graph, canonical	c377a6beff60a9bd35e254139815b2706d7c608e94676792388efbb76d9bb6
	coloring	4e9cb35f9800ea0ffc4189d81ae2aee7e3ddc7156a24e7ea2486fdcf2fda9da
C2000.9	graph, raw	aa4b1d7df7f9c1afb8c9f35cf5bdfc6370fd27db5c9f3fd5409dfd0d164f16
	graph, canonical	a7083cb6110feefea75ed6728fa13968a1533cdc6751717ca09e99b2eafb8bfd
	coloring	4461990a562e2e5616d38709332f3ff1b634753009b97f39627026b6c028d0a2

E.2 What the checkers verify

A coloring is accepted only as a complete assignment of positive labels with distinct labels on every edge; its direct, sorted-class and Ferrers sums are recomputed. The vertex assignments themselves, not their discovery logs, are the upper-bound witnesses.

A residual descriptor specifies the top classes, the uncovered vertices, the eligible sizes, the quotas or equalities, the scores and the target. The eligible columns are regenerated from the graph in increasing-tuple order; for the DSJC250.9 models of Table A.2, the supplied column file must agree exactly with that canonical sequence and its digest. Each tree descriptor binds the model, the columns and the compressed tree, and the certificate collection binds all eleven descriptors, so each dual or tree is checked against the model it is claimed for.

Rational records carry integer numerators and a positive common denominator, and every column inequality and final strict comparison is recomputed in integers. Tree checks reject cycles, shared or orphan records, inactive branch columns, overlap with selected columns, quota underflow and unclosed leaves; both children must be present and every record must be reachable exactly once. The integrated DSJC250.9 checker also accepts leaves closed by an explicit matching and Tutte–Berge barrier (Appendix D), a leaf type its independent checker rejects. Such a closure is valid only when no other positive-score columns remain at the leaf, the pairs carry the assumed score coefficient, and the remaining quotas cannot increase the bound. All six trees used here have only integer-dual leaves (Table A.3), so the difference does not enter the proof.

For DSJC1000.9, the size-5 cap is bound to scenario 111, its three selected 6-sets, the derived residual vertices and the denominator, and the checker verifies the strict threshold 135 before using the cap 134. For C2000.9, the sum-master check verifies the stable-set support, vertex loads, threshold coordinates, unit-cell costs and dual inequalities; its optional constraint $q_1 \geq 390$ selects the second primal solution of [Section 6.2](#).

The DSJC250.9 witness bundle `primal_witnesses.json` used by the integrated checker has SHA-256

```
32d5871ce351a9aee636a2d528ad94c5c9a19481fe941be08df3a12a58e4e7a6.
```

Byte comparisons should use this bundle and the descriptor collection rather than values retyped from this document.

E.3 Replay commands

The commands below use version 1.0.0 of the companion package, available at <https://doi.org/10.5281/zenodo.22952311> and in the repository [certified-sum-coloring-bounds](#). Paths are relative to its root, and its `REPRODUCE.md` documents requirements, manifests and expected outputs. The canonical graph files of [Table E.1](#) are obtained and checked by

```
./scripts/fetch_graphs.sh
```

DSJC250.9 and DSJC500.9. The first command replays the DSJC250.9 proof and profile enumeration. The second checks the archived direct master of [Section 4.6](#): it reconstructs the model, checks its manifest, checks both returned colorings and inspects the recorded solver results, without rerunning the optimization. The third replays the DSJC500.9 certificates.

```
./evidence/dsjc250/public_artifact/scripts/replay_all.sh \
  graphs/DSJC250.9.col build/replay-dsjc250
./evidence/dsjc250/direct_integer_master/verify.sh \
  graphs/DSJC250.9.col
(cd evidence/dsjc500/dsjc500_integrated && ./scripts/replay.sh)
```

The DSJC500.9 replay checks the initial exclusions, the P2/P4 4-set trees, the 17 conditional trees and 30 mappings, the splitting paths of [Appendix C](#) and the certificate of [Appendix D](#), and reconstructs the census of 1,165 cells.

The rational points of [Section 5.4](#) are `P5E_fractional_exact.json`, `P5F_fractional_exact.json` and `P7F_fractional_exact.json`, which correspond in order to the rows of [Table 7](#). They are in

```
evidence/dsjc500/fractional_profile_points/certificates/fractional_
lp/.
```

The associated reconstruction record fixes the order of the stable sets and of the maximum packings. The package's `README.md` records the graph, point, reconstruction and census identities and the scope of its bounded C++20 cross-check, which reconstructs the graph and packing order, checks the rational points in the model ([13](#)), and verifies their correspondence with the three lowest cells. With `CHECK_WORK` set to a new output directory outside the checkout, run

```
bash evidence/dsjc500/fractional_profile_points/verify.sh "$CHECK_WORK"
```

The entry point verifies its input identities before checking the points.

DSJC1000.9. The certificates are replayed by

```
(cd evidence/dsjc1000/DSJC1000.9_EXACT_CONDITIONAL_RELEASE_20260728 \  
  && PYTHONDONTWRITEBYTECODE=1 bash scripts/verify_all.sh . \  
  && sha256sum -c MANIFEST.sha256)
```

The replay regenerates the complete census and checks the coloring, `certificates/full_master_dual.json`, the eight `certificates/conditional_*.json` files and `certificates/size5_packing_cap_111.json`. It also runs tests that malformed certificates are rejected.

C2000.9. The coloring and the top-layer certificate are checked by

```
python3 evidence/c2000_9/audit_tools/verify_c2000_graph_witness.py \  
  --graph evidence/c2000_9/C2000.9.col \  
  --coloring evidence/c2000_9/upper_witness/best.coloring \  
python3 evidence/c2000_9/top_lower_certificate/verify_c2000_9_top_envelope.py \  
  --sets evidence/c2000_9/top_lower_certificate/stable_sets_size_6.tsv.gz \  
  --certificate evidence/c2000_9/top_lower_certificate/top_layer_certificate.json
```

The top certificate uses the complete canonical family of 6-sets, which can be regenerated independently with the census enumerator:

```
mkdir -p build/c2000 \  
g++ -O3 -std=c++20 evidence/c2000_9/audit_tools/census_stable_sets.cpp \  
  -o build/c2000/census_stable_sets \  
build/c2000/census_stable_sets evidence/c2000_9/C2000.9.col 7 \  
build/c2000/size6.tsv
```

The fractional-coloring and sum-master certificates are checked by

```
(cd evidence/c2000_9/c2000_fractional_exact_certificate_20260913T122045Z \  
  && sha256sum -c SHA256SUMS \  
  && PYTHONDONTWRITEBYTECODE=1 python3 source_snapshot/verify_fractional_coloring.py . \  
  && PYTHONDONTWRITEBYTECODE=1 python3 source_snapshot/verify_sum_master.py .)
```

The two primal solutions of [Section 6.2](#) are `EXACT_FRACTIONAL_COLORING.json` and `SUM_MASTER_K390_PRIMAL.json`; the dual is reconstructed by `source_snapshot/verify_sum_master.py` from the certificate's 52-vertex transversal and the slopes given in [Section 6.2](#).

E.4 The archived HiGHS runs

The two one-thread HiGHS 1.11.0 runs of [Section 4.6](#) took 103.558 and 103.374 seconds in the recorded environment. Their raw floating-point objective and bound fields are 8276.9999999999982, displayed as 8,277 with zero gap. The coloring they returned differs from the one in [Table E.1](#) and has SHA-256

7920ddafee753cc3554b24de0071b95f4a837373b604dc9aaf576925f11d3dad.

F Notation

symbol	meaning
λ, ℓ	Sorted class sizes $\lambda_1 \geq \dots \geq \lambda_\ell$ of a coloring with ℓ classes.
h_s, g_t	Number of classes of size exactly s ; Ferrers coordinate $g_t = \sum_{s \geq t} h_s$.
$T(q), \Phi(h)$	$T(q) = q(q+1)/2$; $\Phi(h) = \sum_t T(g_t)$, the cost of profile h .
B	One less than the value of a known coloring; colorings of value at most B are excluded.
$\mathcal{S}, \mathcal{S}_s$	All nonempty stable sets; those of size s .
$x_S, q_t, u_{t,j}, N_t$	Master variables, threshold coordinates $q_t = \sum_{ S \geq t} x_S$, unit cells, and their number $N_t = \lfloor n/t \rfloor$.
$F, E, \mathcal{Q}_F(E)$	Top packing, eligible sizes, and the complete residual universe (Definition 3.1).
n_s, c_s, a_s, τ, p	Residual count, cap, score of size s ; target; score already selected in a tree.
D, y_v, μ_s	Scaling denominator, vertex weights and count multipliers of a dual certificate (Lemma 3.3).
ρ	Radix $ V + 1$ of the lexicographic scores Z_3, Z_2 .
X_i, Pi, A_i	Maximum stable sets of DSJC250.9; maximum 5-set packings of DSJC500.9; stable 6-sets of DSJC1000.9.
$b, k, R_b, \mathcal{S}_b, r_t$	Mask, number of selected 6-sets, residual vertices and universe, and residual threshold coordinates for DSJC1000.9.
$\pi_v, \beta_t, \delta, \Gamma_k, w_v$	Vertex prices, slopes, cap multiplier and slope penalty of a master dual; weights of the 5-set cap.
σ_2, σ_3	Splitting moves of Appendix C.

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